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A waveform-similarity-based protection scheme for the VSC-HVDC transmission lines

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ABSTRACT

The protection for the voltage source converter based high voltage direct current (VSC-HVDC) transmission lines must detect the faulty DC lines and faulty poles within a few milliseconds, because the fault current rises quickly. The traditional protections are easily influenced by fault resistance and signal noise. What is worse, the double-pole-to-ground (DPG) fault with its positive-pole fault resistance unequal to negative-pole fault resistance cannot be detected accurately. To solve the problems, a non-unit protection scheme based on the waveform similarity and wavelet transform modulus maximum (WTMM) theories for the VSC-HVDC lines is proposed in this paper. A four-terminal VSC-HVDC power network is modeled in PSCAD to verify the protection. Extensive simulations of various fault situations prove that the protection scheme can detect high-fault-resistance, DPG, and remote faults. Moreover, it is insensitive to the sampling frequency, signal noise, smoothing reactor, and start delay. Furthermore, it can meet the rapidity requirement. Therefore, the proposed protection can identify the DC faults accurately, which minimizes the disturbance to power grids and improves system security and stability.

1. Introduction

Renewable clean resources such as water, wind, and solar have been widely exploited, which is an effective way to solve the problems of environmental pollution and fossil fuel crisis [1,2]. Moreover, the growing penetration of renewable clean energy has produced substantial changes in the technology and organization of the national energy system [3–5]. However, the resources are usually available in remote regions. The voltage source converter based high voltage direct current (VSC-HVDC) system can transmit large-capacity clean electricity [6–8]. In addition, it has the remarkable advantages of separate power control, lower line losses, and asynchronous interconnections [9–11].

The overhead line is usually used in power transmission with long distance. However, more faults will occur on the overhead transmission line, compared with the cable [12]. To reduce the destructive influence caused by DC faults, the faulty DC lines should be isolated by the DC circuit breakers (DB) as soon as possible and other healthy DC lines should be still in operation. The operation of DBs must be controlled by the DC protection. Therefore, the selective, reliable, and rapid protection with high robustness for VSC-HVDC lines is crucial [13].

The DC line protections include the unit protection (based on double-ended electrical quantity measurements) and the non-unit protection (based on single-ended measurements).

The unit protection has a definite protection boundary but it requires communication. The differential protection relies on a rigorous data synchronization and its reliability is influenced by the transient current of line distributed capacitors [14,15]. The transient protection needs high-fidelity communication and its accuracy is influenced by the fault resistance and signal noise [16,17]. The parameter identification protection cannot select the faulty poles [18]. The pilot protection based on traveling waves (TW) has a high demand for sampling frequency (up to 1 MHz) [19]. In conclusion, due to the dependence on communication between two stations, the unit protection cannot satisfy rapidity requirement for the line main protection, whereas it can act as the line backup protection.

The non-unit protection can identify the faulty line and faulty pole with the local measuring unit, so communication between two stations is not required. The current-derivative transient protection is sensitive to the fault resistance and fault distance [20,21]. The traveling wave protection is sensitive to the signal noise and sampling equipment with high performance is required [22,23]. The undervoltage protection cannot detect double-pole faults including the double-pole-to-ground (DPG) faults and pole-to-pole (PP) faults and the overcurrent protection cannot identify the high-fault-resistance faults [24,25]. In summary, because of the independence of communication, the non-unit

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Nomenclature

U, u	symbol for voltage (kV-kilovolt)
I, i	symbol for current (kA-kiloampere)
P	symbol for power (MW-megawatt)
R, r	symbol for resistance (Ω-ohm)
Z	symbol for impedance (Ω-ohm)
Z_c	symbol for wave impedance (Ω-ohm)
L	symbol for inductance (mH-millihenry)
l	symbol for line length (km-kilometer)
f	symbol for fault component
subscript 0	symbol for 0-mode component
subscript 1	symbol for 1-mode component
subscript p	symbol for positive pole DC line
subscript n	symbol for negative pole DC line
DC	direct current
AC	alternating current
VSC-HVDC	voltage source converter based high voltage direct current
WTMM	wavelet transform modulus maximum
DB	direct current circuit breaker
MU	measuring unit
TW	traveling wave
IFTW	initial fault traveling wave
IFVTW	initial fault voltage traveling wave
IFCTW	initial fault current traveling wave
IFRTW	initial fault reverse traveling wave
PCC	Pearson correlation coefficient
PPG	positive pole to ground
NPG	negative pole to ground
PP	positive pole to negative pole
DPG	double pole to ground

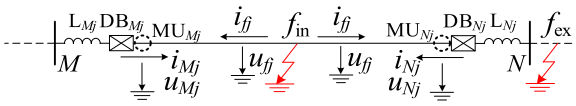


Fig. 1. The IFTWs recorded by measuring units.

protection can satisfy the rapidity requirement.

What is worse, the DPG fault with its positive-pole fault resistance (R_{pf}) unequal to negative-pole fault resistance (R_{nf}) cannot be identified accurately. Because the energy of the faulty-pole fault transient component is more than that of the unfaulty-pole fault transient component, the comparisons (such as differences or ratios) between the energy of double-pole fault transient components (such as the voltage [26], the current [20], and the power [29]) can be used to identify the faulty pole. It is certain that the energy method can detect the positive-pole-to-ground (PPG) faults (the comparisons are greater than a threshold value), the negative-pole-to-ground (NPG) faults (the comparisons are less than a threshold value), and the PP faults (the comparisons are about 1). Also, since the characteristic of DPG faults with R_{pf} equal to R_{nf} is similar to that of PP faults, the kind of DPG fault can be also detected by the method. However, with R_{pf} unequal to R_{nf}, the DPG fault has a very different characteristic, which makes the energy method misjudge (according to 4.3).

Considering the disadvantages of aforementioned protections, the waveform similarity protection method has been deeply researched in [2629] and it is barely influenced by the sampling frequency and signal noise. However, the protection method cannot detect DPG faults (R_{pf} unequal to R_{nf}) and existing waveform similarity protections mostly require communication. Therefore, this paper proposes a non-unit protection scheme for the DC line with waveform similarity and wavelet transform modulus maximum (WTMM) introduced to solve the problems.

This paper has the following contributions:

According to the exponential coefficients of the initial fault voltage traveling waves (IFVTW) mode components derived in the paper, the waveform similarity differences between internal and external faults are proved in theory and the proposed protection can identify all kinds of internal faults.

According to the amplitude comparison between 0-mode and 1-mode initial fault reverse traveling waves (IFRTW) derived in the paper, the characteristic differences among various fault types are proved in theory and the proposed protection can identify the DPG fault (R_{pf} unequal to R_{nf}).

The paper is organized as follows. Section 2 describes the characteristics of initial fault traveling waves (IFTWs) and analyzes their waveform similarity based on the Pearson correlation coefficient (PCC). The protection scheme is given in Section 3. Section 4 shows the PSCAD simulation result. Section 5 draws the conclusion.

2. Basic principle

2.1. Waveform similarity of IFTWs based on PCC

As shown in Fig. 1, in the case of an internal DC fault f_{in}, the measuring unit MU_{Mj} records the initial fault voltage traveling wave (IFVTW) u_{fj} and the initial fault current traveling wave (IFCTW) i_{fj} from f_{in}. j can be p and n, which means the positive pole and negative pole in this paper, respectively. It should be noticed that the following research is based on fault components.

To decouple the coupling components of positive and negative poles, the calculation for their mode components is

$$\begin{bmatrix} u_0 \\ u_1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} u_p \\ u_n \end{bmatrix} \\ \begin{bmatrix} i_0 \\ i_1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} i_p \\ i_n \end{bmatrix} \tag{1}$$

where u_p (i_p) and u_n (i_n) are the positive-pole and negative-pole voltages (currents), respectively; u_0 (i_0) and u_1 (i_1) are the 0-mode and 1-mode voltages (currents) with taking positive-pole voltages (currents) as the references, respectively.

The high frequency component of IFTWs is so faster than the low frequency component that MUs will receive high frequency component first. Besides, the 1-mode component is more stable within the high-frequency band [29]. Therefore, the 1-mode component is selected for waveform similarity.

Moreover, the positive direction of TW is defined from the DC busbar to line. The 1-mode IFTW calculated by (1) between the voltages/ currents at MU_{Mj} (U_{M1}/ I_{M1}) and f_{in} (U_{f1}/ I_{f1}) have the following relational expression in frequency domain (shown in capital letters U and I).

$$\begin{cases} U_{M1}(s) = e^{-\gamma_1(s)x} U_{f1}(s) \\ I_{M1}(s) = e^{-\gamma_1(s)x} I_{f1}(s) \end{cases} \tag{2}$$

where x is the fault distance, and $\exp(-\gamma_1(s)x)$ is the propagation function of DC line, which is given as

$$\begin{cases} e^{-\gamma_1(s)x} \approx \dfrac{K_{a1}}{1 + sT_{a1}} e^{-sT_{w1}} \\ T_{w1} = x/v_1 \end{cases} \tag{3}$$

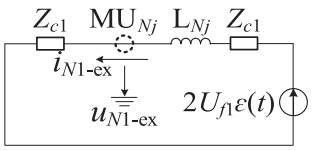


Fig. 2. The 1-mode Peterson equivalent circuit.

where T_{w1} is the wave propagation time from the fault point to MU_{Mj}. v_1 is the traveling wave velocity. $\exp(-sT_{w1})$ represents the time delay of IFTW propagation. $K_{a1}/(1 + sT_{a1})$ describes the attenuation and distortion of IFTW propagation. K_{a1} and T_{a1} are the attenuation and dispersion coefficients, which are given as

$$\begin{cases} K_{a1} = 1 - k_{a1}x \\ T_{a1} = \tau_{a1}x \end{cases} \tag{4}$$

where k_{a1} and τ_{a1} are the attenuation coefficient and dispersion coefficient per unit length, respectively.

So, as shown in Fig. 1, in the case of f_{in}, the 1-mode IFVTW $U_{M1\text{-in}}$ at MU_{Mj} in frequency domain is

$$U_{M1-in}(s) = \frac{K_{a1}}{1+sT_{a1}}\frac{U_{f1}}{s}e^{-sT_{w1}} \tag{5}$$

where U_{f1} is the amplitude of 1-mode IFVTW [30].

According to the inverse Laplace transform and (5), $u_{M1\text{-in}}$ in time domain (shown in lowercase letters u and i) is

$$u_{M1\text{ - in}}(t) = U_{f1}K_{a1}\left[1 - e^{-(t-T_{w1})/T_{a1}}\right]\varepsilon(t - T_{w1}) \tag{6}$$

where $\varepsilon(t)$ is a step input.

According to the positive direction, $i_{M1\text{-in}}$ in time domain is

$$i_{M1\text{ - in}}(t) = -\frac{U_{f1}K_{a1}}{Z_{c1}}\left[1 - e^{-(t-T_{w1})/T_{a1}}\right]\varepsilon(t - T_{w1}) \tag{7}$$

where Z_{c1} is the 1-mode DC line wave impedance.

In addition, the TW can be decomposed into forward TW and reverse TW. The calculation formulae for mode components of initial fault reverse traveling waves (IFRTWs) u_r are

$$\begin{cases} u_{r0} = u_0 - Z_{c0}i_0 \\ u_{r1} = u_1 - Z_{c1}i_1 \end{cases} \tag{8}$$

According to (6) and (7), the IFRTWs $u_{Mr0\text{-in}}$ and $u_{Mr1\text{-in}}$ are

$$\begin{cases} u_{Mr0\text{ - in}}(t) = 2U_{f0}K_{a0}\left[1 - e^{-(t-T_{w0})/T_{a0}}\right]\varepsilon(t - T_{w0}) \\ u_{Mr1\text{ - in}}(t) = 2U_{f1}K_{a1}\left[1 - e^{-(t-T_{w1})/T_{a1}}\right]\varepsilon(t - T_{w1}) \end{cases} \tag{9}$$

where U_{f0} is the amplitude of 0-mode IFVTW [30].

If the attenuation and distortion of IFTW propagation are not considered, (9) means that before the next reverse TW reaches the MU_{Mj}, $u_{Mr0\text{-in}}$ and $u_{Mr1\text{-in}}$ remain always constant at $2U_{f0}$ and $2U_{f1}$, respectively.

In addition, in the case of an external DC fault f_{ex}, assuming that the wave reflected by the L_{Mj} does not reach the MU_{Nj}, the 1-mode Peterson equivalent circuit is shown in Fig. 2.

The 1-mode IFCTW $I_{N1\text{-ex}}$ at MU_{Nj} in frequency domain is

$$I_{N1\text{ - ex}}(s) = \frac{2U_{f1}}{s(2Z_{c1} + sL_{Nj})} \tag{10}$$

The $I_{N1\text{-ex}}$ will propagate to MU_{Mj} along line MN. According to (3), with the attenuation and distortion on the whole line (the total length is l_{MN}), the 1-mode IFCTW $I_{M1\text{-ex}}$ is

$$I_{M1\text{ - ex}}(s) = -\frac{2U_{f1}K_{al1}}{s(2Z_{c1} + sL_{Nj})(1 + sT_{al1})}e^{-sT_{wl1}} = -\frac{2U_{f1}(1 - k_{a1}l_{MN})}{s(2Z_{c1} + sL_{Nj})(1 + s(\tau_{a1}l_{MN}))}e^{-s(l_{MN}/v_1)} \tag{11}$$

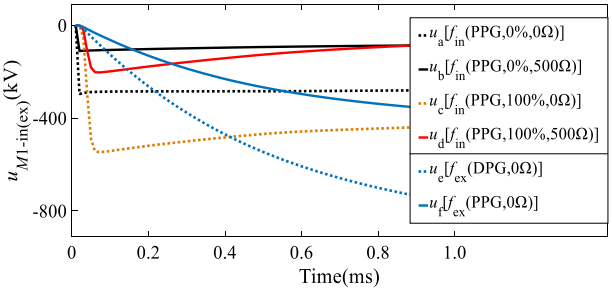


Fig. 3. $u_{M1\text{-in(ex)}}$ at MU_{Mj} with different faults.

According to the inverse Laplace transform and (11), $i_{M1\text{-ex}}$ in time domain is

$$i_{M1\text{ - ex}}(t) = -\frac{U_{f1}K_{al1}}{Z_{c1}}\left(1 - e^{-(t-T_{wl1})/\left(T_{al1}+L_{Nj}/2Z_{c1}\right)}\right)\varepsilon(t - T_{wl1}) \tag{12}$$

According to the positive direction, $u_{M1\text{-ex}}$ in time domain is

$$u_{M1\text{ - ex}}(t) = U_{f1}K_{al1}\left(1 - e^{-(t-T_{wl1})/\left(T_{al1}+L_{Nj}/2Z_{c1}\right)}\right)\varepsilon(t - T_{wl1}) \tag{13}$$

According to (6) and (13), the amplitude $U_{f1}K_{a1}$ ($U_{f1}K_{al1}$) of $u_{M1\text{-in}}$ ($u_{M1\text{-ex}}$) reflects the fault resistance and TW propagation attenuation. Besides, the exponential coefficient of $u_{M1\text{-in}}$ ($u_{M1\text{-ex}}$) reflects the fault distance and TW propagation distortion. According to (4), for internal faults, the more the fault distance is, the larger the dispersion coefficient T_{a1}. But since τ_{a1} is very small, T_{a1} does not change much, and so does the exponential coefficient $1/T_{a1}$ of $u_{M1\text{-in}}$. Therefore, 1-mode IFVTWs at MU_{Mj} between internal faults with various fault distances and fault resistances should be in high similarity.

For external faults, the exponential coefficient satisfies

$$\frac{1}{T_{al1} + L_{Nj}/2Z_{c1}} < \frac{1}{T_{al1}} \leqslant \frac{1}{T_{a1}} \tag{14}$$

The exponential coefficient of $u_{M1\text{-ex}}$ is much smaller than that of $u_{M1\text{-in}}$ due to the smoothing reactor L_{Nj} (wave impedance discontinuity). Thus, the 1-mode IFVTWs between external and internal faults should be greatly different.

As shown in Fig. 1, in the case of internal (external) positive-pole-to-ground (PPG) faults f_{in} (f_{ex}), different $u_{M1\text{-in}}$ ($u_{M1\text{-ex}}$) can be recorded by MU_{Mj}, which is given in Fig. 3. For example, "f_{in} (PPG, 100 %, 500 Ω)" means that the PPG fault occurs at f_{in}, 100 % distance away from MU_{Mj} with fault resistance of 500 Ω.

The Pearson correlation coefficient (PCC) of signals x and y is $c(x, y)$, calculated as

$$c(x,y) = \frac{\sum_{i=1}^{n}\left(x_i - \left(\sum_{i=1}^{n}x_i\right)/n\right)\left(y_i - \left(\sum_{i=1}^{n}y_i\right)/n\right)}{\sqrt{\sum_{i=1}^{n}\left(x_i - \left(\sum_{i=1}^{n}x_i\right)/n\right)^2}\sqrt{\sum_{i=1}^{n}\left(y_i - \left(\sum_{i=1}^{n}y_i\right)/n\right)^2}} \tag{15}$$

where n is the signal sample number. $c(x, y)$ can reflect the directionality and waveform similarity between x and y.

The range of PCC is $-1 \leq c(x, y) \leq 1$. The values of -1 and 1 are completely positive and negative relationship, respectively. The greater positive value indicates a higher similarity between two samples. The value of 0 is a complete difference. The negative value means the two signal samples are not similar because they are in two different directions.

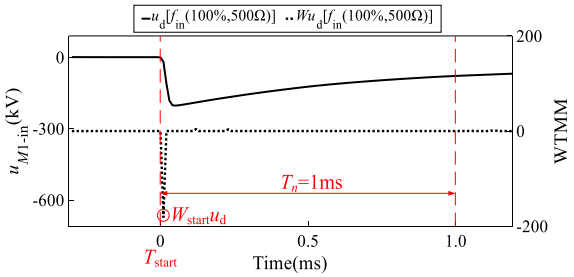


Fig. 4. $u_{M1\text{-in}}$ and MTWW at MU$_{Mj}$ with 100% fault distance.

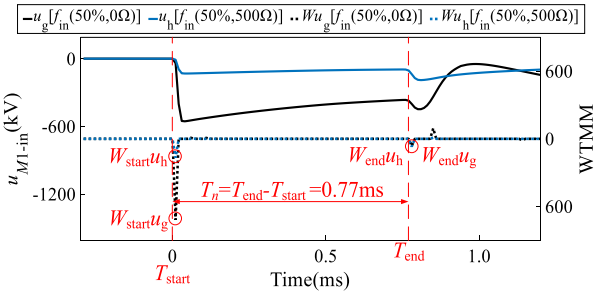


Fig. 5. $u_{M1\text{-in}}$ and MTWW at MU$_{Mj}$ with 50% fault distance.

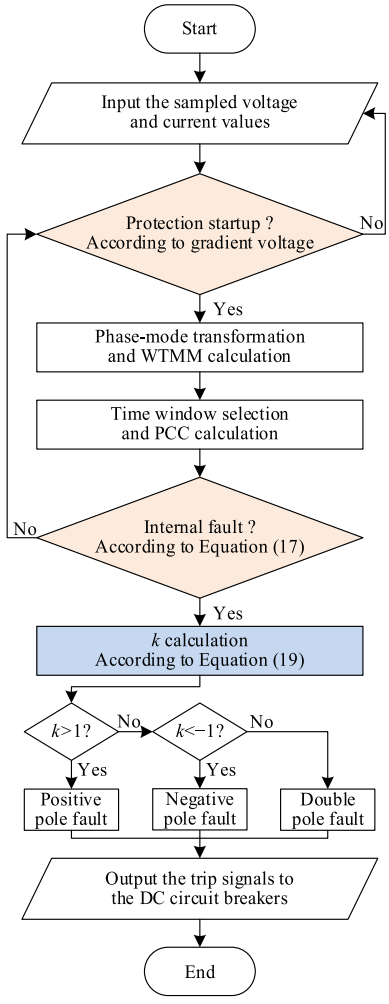


Fig. 6. Flowchart of the proposed protection scheme.

As shown in Fig. 3, there is a high similarity among the 1-mode internal-fault IFVTWs (u_a, u_b, u_c, u_d) whereas there is lower or no similarity between $u_{a/b/c/d}$ and 1-mode external-fault IFVTWs ($u_{e/f}$). So, c ($u_{a/b/c/d}$, $u_{a/b/c/d}$) is much more than c($u_{a/b/c/d}$, $u_{e/f}$), making it more reliable to distinguish between the internal and external faults. Moreover, when the most serious (most remote and highest fault resistance) fault f_{in} (PPG, 100 %, 500 Ω) occurs, c(u_d, u_d) is the maximum, equal to 1, making it more sensitive to this kind of fault. So, u_d can be the Reference Waveform (u_R). The PCC of u_R and 1-mode IFVTWs by MUs can be used to identify an internal fault. The method can be also applied to other DC lines and the 1-mode IFVTWs with the most serious internal PPG fault can be selected as the Reference Waveform.

2.2. Time window length based on WTMM

Due to its shift-invariance with time, the wavelet transform modulus maximum (WTMM) can be used to detect the signal singularity. Firstly, the dyadic wavelet transform is used to decompose the discrete traveling wave $TW(n)$ with the initial frequency band [0, f_i]. It can be disintegrated into two parts including wavelet approximations $WA(2^1)TW(n)$ with the frequency band [0, $f_i/2$] and wavelet details $WD(2^1)TW(n)$ with the frequency band [$f_i/2$, f_i] in the first scale. Secondly, decompose the $WA(2^1)TW(n)$ into the corresponding $WA(2^2)TW(n)$ and $WD(2^2)TW(n)$ in the second scale. And repeat the decomposition in various scales. Thirdly, calculate the local maxima of the corresponding wavelet details in different scales. So, these maxima are defined as the WTMM of $TW(n)$. Because the maxima are one-to-one corresponding to the TW singularity points, WTMM is employed to detect the start and end time of 1-mode IFVTWs [31], which is expressed as

$$|Wu_{M1}| \geqslant W_{set} \tag{16}$$

where Wu_{M1} is the WTMM of 1-mode IFVTW and W_{set} is the setting value. The time window length is T_n, with the maximum being 1 ms. And the number of sampling data is n, equal to T_n multiplied by sampling frequency.

The first Wu_{M1} satisfying the criterion is $W_{start}u_{M1}$, with the corresponding sampling time of Wu_{M1} prior to $W_{start}u_{M1}$ denoted by T_{start}. Then the 1-mode IFVTW data will be sampled in the next 1 ms. In this period, if there is not another Wu_{M1} that can satisfy the criterion, the time window length is T_n = 1 ms. But if there is the second Wu_{M1} ($W_{end}u_{M1}$) that can satisfy the criterion, it means that the reflected wave arrives at MU$_{Mj}$, with corresponding sampling time of Wu_{M1} prior to $W_{end}u_{M1}$ denoted by T_{end}. So, it is only between the T_{start} and T_{end} that TW is the 1-mode IFVTW. In this case, the variable time window length is $T_n = T_{end} - T_{start}$.

To make the selection principle of time window length easily understood, Fig. 4 and Fig. 5 show the $u_{M1\text{-in}}$ and WTMM, when the internal PPG faults f_{in} (shown in Fig. 1) occur. According to numerous simulation tests, the WTMM in the second scale is selected to calculate the time window length. As shown in Fig. 4, for f_{in} (100 %, 500 Ω), the time window length T_n is 1 ms, because there is not another Wu_d satisfying the criterion. As shown in Fig. 5, for f_{in} (50 %, 0/500 Ω), the time window length T_n is 0.77 ms, because there is the second $Wu_{g/h}$ satisfying it.

3. Protection scheme

The proposed protection scheme consists of three units, which are the start-up unit, internal fault identification unit, and faulty pole detection unit. Moreover, the flowchart of the proposed protection is shown in Fig. 6.

Firstly, if the protection start-up criterion based on gradient voltages is satisfied, the protection start-up signal will be given out [26].

Secondly, the waveform similarity based on WTMM and PCC can be employed to identify internal DC line faults. According to 2.1, the in-

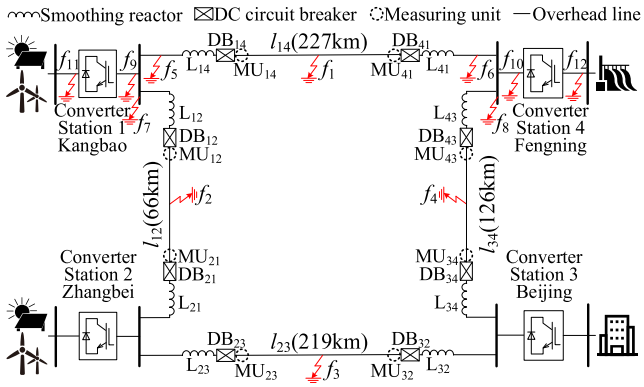


Fig. 7. The VSC-HVDC transmission system.

Table 1
Test results of external faults.

$c(u_R, u_{141})$	PPG	NPG	DPG	PP	O/TPG
f_2	−0.6476	−0.6472	−0.6324	−0.6328	/
f_3	−0.6815	−0.6821	−0.6830	−0.6836	/
f_4	−0.6105	−0.6112	−0.5930	−0.5921	/
f_5	−0.5884	−0.5885	−0.6071	−0.6073	/
f_6	−0.5688	−0.5698	−0.5763	−0.5762	/
$f_{7/9}$	−0.5887	−0.5889	−0.6069	−0.6067	/
$f_{8/10}$	−0.5649	−0.5663	−0.5721	−0.5718	/
f_{11} and f_{12}	/	/	/	/	SNS

ternal fault identification criterion is

$$c(u_R, u_1) \rangle c_{set} \tag{17}$$

where $c(u_R, u_1)$ is the PCC of the Reference Waveform (u_R) and 1-mode IFVTWs. c_{set} is the threshold value and it can be set as 0, which is to determine whether $c(u_R, u_1)$ is positive or not. If the $c(u_R, u_1)$ is positive, it means that there is a similarity between u_R and the 1-mode IFVTWs recorded by MUs. Also, it indicates that an internal fault is identified.

Thirdly, the mode components of IFRTWs recorded by MUs can be employed to detect DC faulty poles. According to 2.1, the ratios of $u_{r0\text{-}in}$ and $u_{r1\text{-}in}$ are

$$\frac{u_{r0\text{-}in}}{u_{r1\text{-}in}} \approx \frac{U_{f0}}{U_{f1}} = \begin{cases} \text{PPG fault: } \dfrac{Z_0}{Z_1} > 1 \\ \text{NPG fault: } -\dfrac{Z_0}{Z_1} < -1 \\ \text{PP fault: } 0 \\ \text{DPG fault: } -1 \leqslant \dfrac{(R_{nf} - R_{pf})Z_0}{(Z_0 + R_{pf} + R_{nf})Z_1} \leqslant 1 \end{cases} \tag{18}$$

where R_{pf}, R_{nf}, and R_f are the fault resistance with various fault types, which include PPG, NPG, PP, and DPG faults. The distinguishment between PP and DPG faults is not required, because they are both double-pole faults and the double-pole DC lines both need deenergizing by DBs. Z_0 and Z_1 are the DC line impedances, and Z_0 is greater than Z_1. So, the faulty pole detection criterion is

$$k = \sum_{t=T_{0start}}^{T_{0end}} u_{r0\text{-}in}(t) / \sum_{t=T_{1start}}^{T_{1end}} u_{r1\text{-}in}(t) \tag{19}$$

where T_{0start} and T_{0end} represent the start and end time of 0-mode IFRTWs, and T_{1start} and T_{1end} represent the start and end time of 1-mode IFRTWs, respectively. The determination of these time and IFRTWs duration are the same as the principle in 2.2. If $k > 1$, it means the positive pole is faulty; if $k < -1$, it means the negative pole is faulty; if $1 \leq k \leq -1$, it means the double poles are both faulty.

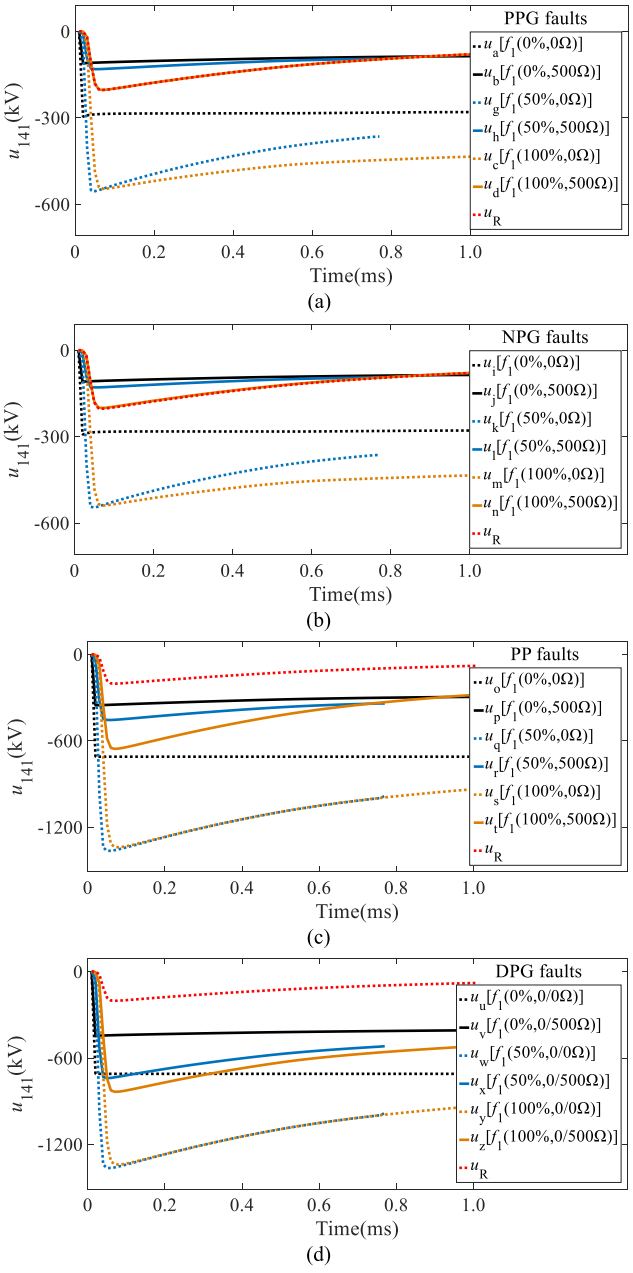


Fig. 8. The Reference Wave (u_R) and u_{141} with different faults. (a) PPG faults. (b) NPG faults. (c) PP faults. (d) DPG faults.

4. Simulation and discussion

To verify the validity of the proposed protection, the Zhangbei four-terminal ± 500 kV VSC-HVDC simulation [26] shown in Fig. 7 is modeled in PSCAD. As shown in Fig. 7, converter stations are connected through overhead bipolar DC transmission lines. For convenience, DC line l_{14} (including positive-pole and negative-pole lines) is taken as the protected line. Without a loss of generality, the verification is focused on the electrical quantities recorded by MU_{14} on the line l_{14}.

The internal fault (f_1) is located in the protection area of line l_{14}, which is between the smoothing reactors L_{14} and L_{41}. And the external faults are situated outside the protection area of line l_{14}, including several types such as the faults on the adjacent DC lines l_{12} (f_2), l_{23} (f_3), and l_{34} (f_4), DC busbar faults (f_7 and f_8), faults (f_5, f_6, f_9, and f_{10}) nearby DC busbars, and AC faults (f_{11} and f_{12}).

For example, "f_1 (DPG, 50 %, 0/500 Ω)" means the DPG fault occurs

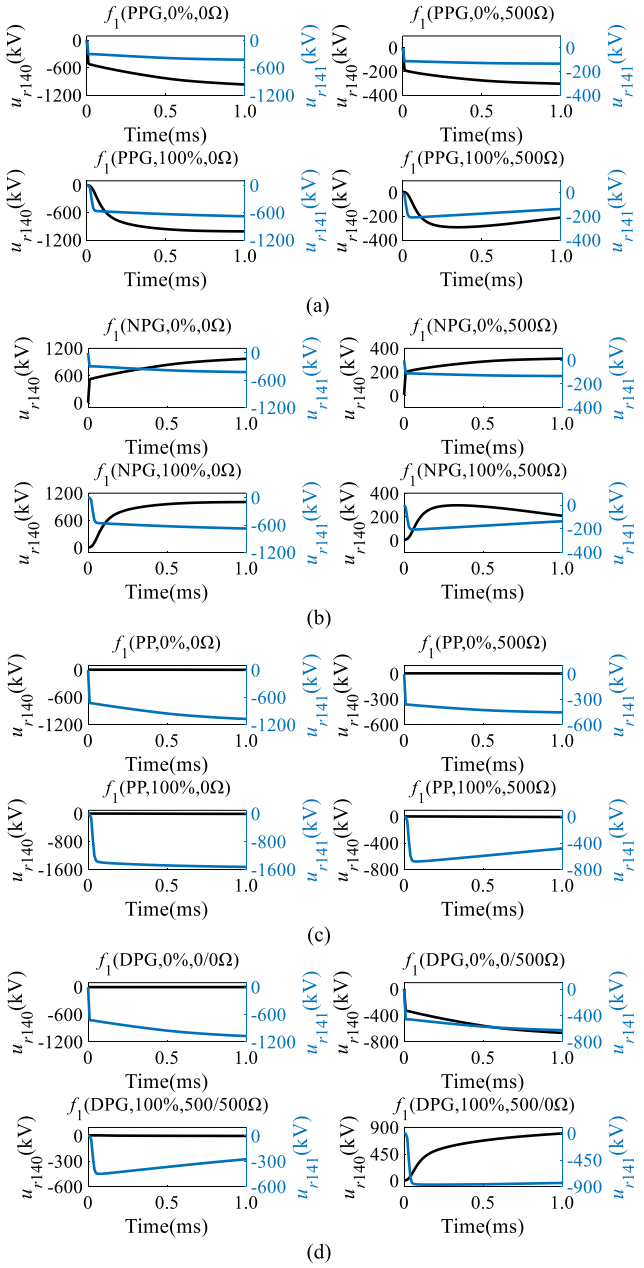


Fig. 9. The 0(1)-mode IFRTWs $u_{r140(1)}$ with different faults. (a) PPG faults. (b) NPG faults. (c) PP faults. (d) DPG faults.

Table 2
PCC of reference waveform and 1-mode IFVTWs.

PPG	u_a	u_b	u_g	u_h	u_c	u_d
u_R	0.3260	0.6369	0.8927	0.8777	0.7839	1.0000
NPG	u_i	u_j	u_k	u_l	u_m	u_n
u_R	0.3196	0.6265	0.8903	0.8742	0.7708	0.9998
PP	u_o	u_p	u_q	u_r	u_s	u_t
u_R	0.3179	0.5925	0.8819	0.8717	0.8883	0.9991
DPG	u_u	u_v	u_w	u_x	u_y	u_z
u_R	0.3179	0.4523	0.8819	0.8869	0.8887	0.9458

Table 3
Test results of k.

x	0 %		50 %		100 %	
R_f	0 Ω	500 Ω	0 Ω	500 Ω	0 Ω	500 Ω
PPG	2.1443	2.1116	1.7855	1.6957	1.4018	1.4070
NPG	−2.1805	−2.2265	−1.8071	−1.7681	−1.4141	−1.4484
PP	−0.0074	−0.0166	−0.0049	−0.0112	−0.0023	−0.0059
R_{pf} / R_{nf}	0/0Ω	0/500 Ω	0/500 Ω	500/0Ω	500/500 Ω	500/0Ω
DPG	−0.0015	0.6670	0.6360	−0.6390	−0.0025	−0.5148

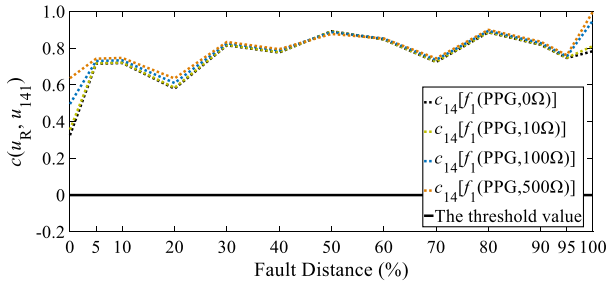


Fig. 10. The results of $c(u_R, u_{141})$ with various fault distances and fault resistances.

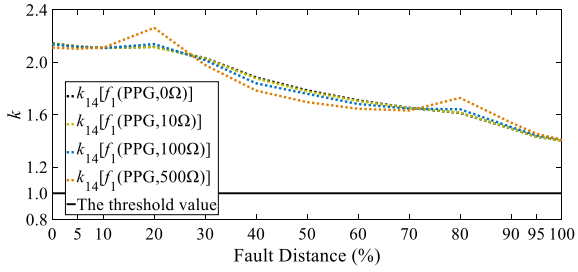


Fig. 11. The results of k with various fault distances and fault resistances.

Table 4
Test results of NPG faults.

Fault resistance	Criterion	12 %	15 %	18 %	21 %
0 Ω	$c(u_R, u_{141})$	0.5350	0.6327	0.7293	0.7773
	k	−2.1100	−2.1786	−2.1341	−2.1154
10 Ω	$c(u_R, u_{141})$	0.5383	0.6386	0.7321	0.7793
	k	−2.1082	−2.1818	−2.1278	−2.1194
100 Ω	$c(u_R, u_{141})$	0.5568	0.6686	0.7466	0.7896
	k	−2.1116	−2.2106	−2.1222	−2.1580
500 Ω	$c(u_R, u_{141})$	0.5748	0.6945	0.7606	0.8001
	k	−2.1314	−2.2789	−2.2121	−2.1876

Table 5
Test results of DPG faults.

R_{pf}	R_{nf}	Criterion	0 %	50 %	100 %
10 Ω	200 Ω	$c(u_R, u_{141})$	0.4492	0.8804	0.9584
		k	0.3408	0.3425	0.2797
200 Ω	10 Ω	$c(u_R, u_{141})$	0.4462	0.8790	0.9577
		k	−0.3475	−0.3430	−0.2774
0 Ω	0 Ω	$c(u_R, u_{141})$	0.3179	0.8819	0.8887
		k	−0.0015	−0.0018	−0.0019
500 Ω	500 Ω	$c(u_R, u_{141})$	0.6415	0.8704	0.9990
		k	−0.0136	−0.0054	−0.0025

Table 6
Test results of various sampling frequencies.

Sampling frequency	DC Fault	Criterion		Result
		$c(u_R, u_{141})$	k	
10 kHz	f_1 (50 %,0Ω)	0.9770	1.6760	Internal PPG
	f_8 (PPG,0Ω)	−0.4759	/	External fault
20 kHz	f_1 (50 %,0Ω)	0.5857	1.7484	Internal PPG
	f_8 (PPG,0Ω)	−0.4495	/	External fault
50 kHz	f_1 (50 %,0Ω)	0.9581	1.7893	Internal PPG
	f_8 (PPG,0Ω)	−0.5101	/	External fault
100 kHz	f_1 (50 %,0Ω)	0.8927	1.7855	Internal PPG
	f_8 (PPG,0Ω)	−0.5649	/	External fault

Table 7
Test results of various smoothing reactors.

Smoothing reactor	DC Fault	Criterion		Result
		$c(u_R, u_{141})$	k	
10mH	f_1 (50 %,500 Ω)	0.8106	1.8222	Internal PPG
	f_7 (DPG,0Ω)	−0.4261	/	External fault
100mH	f_1 (50 %,500 Ω)	0.8602	1.7044	Internal PPG
	f_7 (DPG,0Ω)	−0.6775	/	External fault
200mH	f_1 (50 %,500 Ω)	0.8777	1.6957	Internal PPG
	f_7 (DPG,0Ω)	−0.6069	/	External fault
300mH	f_1 (50 %,500 Ω)	0.8837	1.6663	Internal PPG
	f_7 (DPG,0Ω)	−0.5266	/	External fault

Table 8
Test results of various signal noise levels.

Signal noise	DC Fault	Criterion		Result
		$c(u_R, u_{141})$	k	
20 dB	f_1 (50 %,0Ω)	0.7877	1.7369	Internal PPG
	f_7 (PPG,0Ω)	−0.4383	/	External fault
30 dB	f_1 (50 %,0Ω)	0.8793	1.7957	Internal PPG
	f_7 (PPG,0Ω)	−0.5516	/	External fault
40 dB	f_1 (50 %,0Ω)	0.8930	1.7839	Internal PPG
	f_7 (PPG,0Ω)	−0.5878	/	External fault
50 dB	f_1 (50 %,0Ω)	0.8931	1.7845	Internal PPG
	f_7 (PPG,0Ω)	−0.5887	/	External fault

Table 9
Test results of various start delay steps.

Start delay	DC Fault	Criterion		Result
		$c(u_R, u_{141})$	k	
0 step	f_1 (50 %,500 Ω)	0.8777	1.6957	Internal PPG
	f_7 (DPG,0Ω)	−0.6069	/	External fault
3 steps	f_1 (50 %,500 Ω)	0.3493	1.7244	Internal PPG
	f_7 (DPG,0Ω)	−0.6083	/	External fault
6 steps	f_1 (50 %,500 Ω)	0.3711	1.7550	Internal PPG
	f_7 (DPG,0Ω)	−0.6081	/	External fault
9 steps	f_1 (50 %,500 Ω)	0.3880	1.7766	Internal PPG
	f_7 (DPG,0Ω)	−0.6081	/	External fault

at f_1, 50 % distance away from MU_{14} with R_{pf} and R_{nf} of 0 Ω and 500 Ω, respectively. In addition, "u_{141}" represents the 1-mode IFVTWs recorded by the MU_{14}. "$u_{r140(1)}$" represents the 0(1)-mode IFRTWs recorded by the MU_{14}.

4.1. Simulation verification

Firstly, the identification of external faults by the proposed protection is verified.

Table 1 lists the test results of external metallic short circuit faults. "SNS" represents the protection start-up is not satisfied. "O/TPG" is the one-phase-to-ground or three-phase-to-ground AC fault. As shown in Table 1, every value is much less than c_{set}, which indicates all external faults are identified.

Secondly, the identification of internal faults by the proposed protection is verified.

The waveforms of u_R and 1-mode IFVTW (u_{141}) are shown in Fig. 8. As shown in Fig. 8, the traveling wave directions and shapes of all u_{141} and u_R are identical.

After the internal fault is identified, it is crucial to detect the faulty pole.

The waveforms of 0(1)-mode IFRTW ($u_{r140(1)}$) recorded by MUs are shown in Fig. 9. As shown in Fig. 9, for PPG and NPG faults, the absolute value of amplitude of u_{r140} is more than that of u_{r141}. And the polarities of u_{r140} and u_{r141} are the same for PPG faults, whereas those are opposite for NPG faults. For PP faults, the amplitude of u_{r140} is very close to 0, and the polarity of u_{r141} is negative. For DPG faults, there are three situations: 1) when R_{pf} is equal to R_{nf}, it is similar to the PP fault; 2) when R_{pf} is greater than R_{nf}, the absolute value of amplitude of u_{r140} is less than that of u_{r141}, and the polarities of u_{r140} and u_{r141} are opposite; 3) when R_{pf} is smaller than R_{nf}, the absolute value of amplitude of u_{r140} is less than that of u_{r141}, and the polarities of u_{r140} and u_{r141} are the same.

Finally, the test results of $c(u_R, u_{141})$ and k are given to verify the criterion reliability and sensitivity.

Table 2 lists the results of $c(u_R, u_{141})$. As shown in Table 2, every value is much greater than c_{set}, which means all internal faults are identified. Furthermore, the minimum $c(u_{o/u}, u_R) = 0.3179$ is still more than 0. This means when the most extreme fault occurs, the PCC criterion can be still fulfilled.

Table 3 lists the results of k. As shown in Table 3, for PPG faults, the minimum value 1.4018 is still much more than 1; for DPG faults, the maximum value 0.6670 is still much smaller than 1. This indicates when the most extreme fault occurs, the k criterion can be still satisfied.

Table 10
Test results of various measurement errors.

Measurement error (%)				DC Fault	Criterion		Result
u_p	u_n	i_p	i_n		$c(u_R, u_{141})$	k	
+10	+10	+10	+10	f_1	0.8737	1.7859	Internal
		+10	−10	(50 %,0Ω)	0.8737	1.7762	PPG
		−10	+10		0.8737	1.7506	
		−10	−10		0.8737	1.7414	
+10	−10	+10	+10	f_1	0.8625	1.6506	Internal
		+10	−10	(50 %,500	0.8625	1.6534	PPG
		−10	+10	Ω)	0.8625	1.6167	
		−10	−10		0.8625	1.6166	
−10	+10	+10	+10	f_7	−0.6069	/	External
		+10	−10	(DPG,0Ω)	−0.6069		fault
		−10	+10		−0.6069		
		−10	−10		−0.6069		
−10	−10	+10	+10	f_8	−0.5721	/	External
		+10	−10	(DPG,0Ω)	−0.5721		fault
		−10	+10		−0.5721		
		−10	−10		−0.5721		

4.2. Performance analysis

The analysis of significant elements which influence the proposed protection performance is given in the following order. These factors include fault distance, fault resistance, sampling frequency, smoothing reactor, signal noise, start delay, measurement error, operational mode, and operating time.

Fault distance and fault resistance are key factors to evaluate the performance of protections. The different PPG faults are simulated, with the test results presented in Fig. 10 and Fig. 11. As shown in Fig. 10 and Fig. 11, the $c(u_R, u_{141})$ and k are insensitive to the fault resistance. Although they change with the increase of fault distance, the $c(u_R, u_{141})$ and k always exceed their threshold values, which means the PPG faults

Table 11
Test results of various operational modes.

Operational mode	DC Fault	Criterion		Result
		$c(u_R, u_{141})$	k	
DC line l_{23} is out of operation	f_1 (50 %,0Ω)	0.8736	1.7982	Internal PPG
	f_1 (50 %,500 Ω)	0.8577	1.6849	Internal PPG
	f_7 (PPG,0Ω)	−0.5878	/	External fault
	f_8 (DPG,0Ω)	−0.5650	/	External fault
Converter station 1 is out of operation	f_1 (50 %,0Ω)	0.8914	1.7804	Internal PPG
	f_1 (50 %,500 Ω)	0.8800	1.6826	Internal PPG
	f_7 (PPG,0Ω)	−0.5879	/	External fault
	f_8 (DPG,0Ω)	−0.5637	/	External fault

Table 12
Test results of the faulty pole detection.

Internal DPG fault		K_1/Result	K_2/Result	k/Result
R_{pf} (Ω)	R_{nf} (Ω)	In [28]	In [20]	In this paper
0	0	1.00/Bipolar fault	0.0012/Bipolar fault	−0.0018/ Bipolar fault
0	500	4.33/Positive pole fault	0.0521/Positive pole fault	0.6360/ Bipolar fault
500	0	0.23/Negative pole fault	−0.0519/Negative pole fault	−0.6390/ Bipolar fault

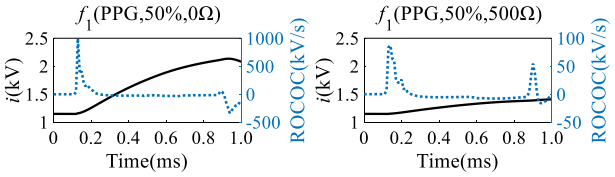


Fig. 12. i_{14} at MU_{14} with its ROCOC.

are detected. Moreover, Table 4 lists the test results of close-by NPG faults. As shown in Table 4, the $c(u_R, u_{141})$ exceeds c_{set}, and the k is less than −1, which means the NPG faults are detected. Besides, Table 5 lists the test results of DPG faults. As shown in Table 5, the $c(u_R, u_{141})$ is more than c_{set}, and the k is between 1 and −1, which means the DPG faults are detected. So, the proposed protection scheme can identify all DC faults and faulty poles correctly, not affected by fault distance and resistance.

To analyse the influence of sampling rates on the protection, the DC voltages and currents are sampled at various sampling frequencies, with the test results listed in Table 6. As shown in Table 6, for the fault f_1, $c(u_R, u_{141})$ and k both exceed the set value. For the faults f_7 and f_8, $c(u_R, u_{141})$ is smaller than c_{set}. So, the proposed protection scheme is not affected by the sampling frequency, because it does not change the trend of IFVTWs and amplitude comparison between 0-mode and 1-mode IFRTWs.

To investigate the impact of smoothing reactor on the method, DC line faults are simulated with different values of L_{14} and L_{41}, with the test results listed in Table 7. As shown in Table 7, for the fault f_1, $c(u_R, u_{141})$ is all close to each other and greater than c_{set}. k decreases slightly as the smoothing reactor increases but it is more than the threshold value. For the fault f_7, $c(u_R, u_{141})$ changes slightly as the smoothing reactor changes but it is still smaller than c_{set}. Therefore, the proposed protection scheme is not subjected to the value of smoothing reactor.

To analyze the impact of signal noise on the method, DC line faults are simulated with different levels of Gauss white noise, with the test results listed in Table 8. As shown in Table 8, for the fault f_1, $c(u_R, u_{141})$ declines slightly as the signal noise level increases but it is more than c_{set}. k is all close to each other and is greater than the setting value. For the fault f_7, $c(u_R, u_{141})$ increases slightly as the noise level increases but it still does not exceed c_{set}. Consequently, the proposed protection scheme has a strong anti-noise capacity, because the noise does not change the polarity, direction, and amplitude comparison of the TWs.

Table 13
The comparisons of the proposed protection and the others.

	Proposed protection	Ref. [19]	Ref. [20]	Ref. [21]	Ref. [22]	Ref. [27]	Ref. [28]
Fault resistance withstand ability	**500 Ω**	300 Ω	100 Ω	10 Ω	**500 Ω**	**500 Ω**	300 Ω
Sampling frequency	100 kHz	1 MHz	**10 kHz**	100 kHz	**10 kHz**	**10 kHz**	**10 kHz**
Can it withstand signal noise?	**Yes**	/	/	**Yes**	No	**Yes**	**Yes**
Can DPG faults be detected?	**Yes**	No	No	No	No	No	No

The start delay of several sampling steps might exist. Thus, the IFVTWs and u_R may not be strictly aligned. To investigate the influence of start delays on the method, DC line faults are simulated with different steps of start delay. One start delay step is equal to one second divided by the sampling frequency. Table 9 lists the test results. As shown in Table 9, for the fault f_1, $c(u_R, u_{141})$ goes down as the delay step goes up but it is greater than c_{set}. k is all close to each other and greater than the setting value. For the fault f_7, $c(u_R, u_{141})$ is all close to each other but is still below c_{set}. Therefore, the proposed protection scheme can make accurate judgements not affected by the start delay.

The measurement error of the measuring device is also crucial to investigate the performance of protections. Taking the Chinese standards GB/T 26216 and GB/T 26217 on electrical measurements as an example, the peak instantaneous errors of DC voltage (u_p and u_n) and current (i_p and i_n) measurements are no more than ± 10 % [27]. DC line faults are simulated with various combinations of measurement errors, with the test results listed in Table 10. As shown in Table 10, the $c(u_R, u_{141})$ and k are both insensitive to the measurement error. It is notable that u_{141} is calculated by positive and negative voltages, so $c(u_R, u_{141})$ is not related to currents and constant with the same voltage measurement error. For the faults f_1 and $f_{7/8}$, $c(u_R, u_{141})$ is much more than and less than c_{set}, respectively. In addition, $c(u_R, u_{141})$ is close to the one without measurement error. And k changes slightly with various errors but it is greater than the setting value. So, the proposed protection scheme is still

accurate in spite of the measurement error of DC voltage and current.

Different DC faults with various kinds of operational modes are considered, with the test results listed in Table 11. As shown in Table 11, for the faults f_1 and f_7, the values of $c(u_R, u_{141})$ and k are approximate to each other respectively, which indicates the faults are all detected. Thus, the operational mode exerts no influence on the protection scheme.

The operating time comprises the communication delay, time window, and computation time. Since the protection detects DC faults only with local measuring units, the communication delay is almost 0 ms. Additionally, the maximal time window is 1 ms. The proposed algorithm consists of 20299 additions and 6554 multiplications. According to the capability of common digital signal processor, the computation time is estimated to be 0.119 ms [32]. To sum up, the operating time is less than 1.119 ms. Thus, it can satisfy the rapidity requirement that the protection needs to identify the faulty DC line correctly within 2 ms in the 500 kV VSC-HVDC transmission system.

4.3. Comparative discussion

The proposed protection scheme is compared with some similar protection methods such as TW-based [19,22] and similarity-based [27,28] schemes and with some different methods such as transient-based [20,21] schemes.

Firstly, the comparison with TW-based protection is discussed.

For the TW protection proposed in [19], it can detect internal metallic DC faults correctly. But the internal fault identification parameter E_R and E_I (both negative values) increase a lot as fault resistance increases. So, for the high-fault-resistance fault, the parameters are extremely close to the setting value 0 and even likely to exceed it, which leads to an incorrect judgement.

In addition, the method using the Teager energy operator (TEO) of TW is given in [22]. For the external fault f_7 (PPG, 0 Ω) with 20 dB Gauss white noise superimposed on the electrical measurement, the calculated S_{TEO} is 6018 (more than the threshold value 5000), which will be identified as an internal fault by mistake, which means the protection is sensitive to the signal noise.

Secondly, the comparison with similarity-based protection is discussed.

For the waveform similarity protections proposed in [27] and [28], they can identify internal DC faults accurately but communication between two converter stations is required. Thus, it cannot satisfy the rapidity requirement for the DC line main protection, but can be used as the DC line backup protection. However, the ratio (K_1) of integration of positive and negative voltage fault components is used in [28] to detect the faulty pole. For the internal fault f_1 (DPG, 50 %), Table 12 lists the test results of K_1 in [28] and k in this paper. This means that the K_1 method can detect DPG faults ($R_{pf} = R_{nf}$) because it has the same characteristic as PP faults, but it will misjudge the DPG faults ($R_{pf} \neq R_{nf}$). What is worse, a faulty pole selection method is not given in [27].

Thirdly, the comparison with transient-based protection is discussed.

The method using the high-frequency components of positive and negative pole line transient currents (Mag_{pos} and Mag_{neg}) is given in [20]. For the internal fault f_1 (DPG, 50 %), Table 12 lists the test results of $K2$ (equal to the magnitude difference between the positive and negative current spectra) in [20] and k in this paper. This indicates that DPG faults ($R_{pf} \neq R_{nf}$) cannot be detected by [20] whereas they can be identified by the proposed protection accurately.

Additionally, the method using the rate of change of current (ROCOC) is given in [21]. For the internal fault f_1 (PPG, 50 %, 0/500 Ω), the measured i_{14} with its ROCOC are shown in Fig. 12. The internal fault identification threshold is 86 kV/s. As shown in Fig. 12, for the low-fault-resistance fault, the ROCOC of i_{14} is much greater than 86 kV/s. But for the high-fault-resistance fault, the ROCOC is less than 86 kV/s, which means the protection cannot identify the internal high-fault-resistance fault.

In conclusion, the comparisons between the proposed method and other protections are given in Table 13. Although the protection needs a higher sampling frequency, it does not rely on the high sampling frequency according to Table 6. The purpose of higher sampling frequency is to sample more voltage and current signals within a fixed time window, which can decrease the adverse impact of missed or false data on the proposed protection and improve the protection reliability. Furthermore, it has a significant superiority in identifying the DPG and high-fault-resistance faults with strong robustness against noise over the other protections.

5. Conclusion

Identification and isolation of the faulty DC lines and poles during the initial short period after a fault is a crucial challenge in the VSC-HVDC transmission system. This paper proposes a non-unit protection scheme based on the PCC and WTMM for the DC lines. In the novel protection method, gradient voltages are used in the protection startup, with waveform similarities of 1-mode IFVTWs and mode components of IFRTWs used to identify the internal fault and detect the faulty pole, respectively.

Numerous simulation tests prove that the protection has great performance. It can identify the high-fault-resistance, close-by, remote, and DPG faults with different fault types. Furthermore, it is insensitive to the sampling frequency, value of smoothing reactor, level of signal noise, start delay, and operational mode. Additionally, it does not need data synchronization. Therefore, the protection has a solid theoretical basis, simple programmed algorithm, clearing threshold criterion, suitable operating time, high judgement accuracy, and strong robustness.

In conclusion, the proposed protection has good adaptability for promotion and application to the main protection for VSC-HVDC lines. Moreover, it can improve the system security.

CRediT authorship contribution statement

Yuhan Zhang: Methodology, Software, Validation, Formal analysis, Data curation, Writing – original draft, Visualization. **Shunliang Wang:** Conceptualization, Investigation, Resources, Supervision, Writing – review & editing. **Tianqi Liu:** Conceptualization, Supervision, Project administration, Resources, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

See Table A1.

Table A1
The main simulation system parameters.

Converter station	Nominal power	Arm inductance	Cell DC capacitor	Number of cells	Smoothing reactor	Leakage reactance	Transformer nominal voltage	Transformer winding	Control mode
1	1500 MW	100mH	10mF	250	200mH	0.15	230 kV/290 kV	YgD-11	P control
2	1500 MW	50mH	10mF	250	200mH	0.15	230 kV/290 kV	YgD-11	P control
3	3000 MW	50mH	15mF	200	200mH	0.15	525 kV/290 kV	YgD-11	V control
4	1500 MW	100mH	10mF	250	200mH	0.15	525 kV/290 kV	YgD-11	P control

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